

fn sample_bernoulli_rational

Michael Shoemate (@Shoeboxam)

May 6, 2026

This proof resides in “**contrib**” because it has not completed the vetting process.

Warning 1 (Code is not constant-time). `sample_bernoulli_rational` takes in an optional `trials` parameter to denote the number of trials to run. The current implementation does not guard against other types of timing side-channels that can break differential privacy, e.g., non-constant time code execution due to branching.

PR History

- [Pull Request #473](#)

This document proves that the implementations of `sample_bernoulli_rational` in `mod.rs` at commit [f5bb719](#) (outdated¹) satisfies its proof definition.

At a high level, `sample_bernoulli` considers the binary expansion of `prob` into an infinite sequence `a_i`, like so: $\text{prob} = \sum_{i=0}^{\infty} \frac{a_i}{2^{i+1}}$. The algorithm samples $I \sim \text{Geom}(0.5)$ using an internal function `sample_geometric_buffer`, then returns `a_I`.

0.1 Hoare Triple

Precondition

Compiler-verified

- Variable `prob` must be of type `T`.
- Variable `constant_time` must be of type `bool`.
- Type `T` has trait `Float`. `Float` implies there exists an associated type `T::Bits` (defined in `FloatBits`) that captures the underlying bit representation of `T`.
- Type `T::Bits` has traits `PartialOrd` and `ExactIntCast<usize>`.
- Type `usize` has trait `ExactIntCast<T::Bits>`.

Caller-verified

None

¹See new changes with `git diff f5bb719..9205d4f rust/src/traits/samplers/bernoulli/mod.rs`

Pseudocode

```
1 # returns a single bit with some probability of success
2 def sample_bernoulli_rational(prob: RBig, trials: Optional[int]) -> bool:
3     numer, denom = prob.into_parts()
4     return numer > UBig.sample_uniform_int_below(denom, trials)
```

Postcondition

Theorem 0.1. For any setting of the input parameters **prob** of type $\mathbb{T}$ restricted to $[0, 1]$, and optionally **trials** of type `usize`, `sample_bernoulli_rational` either

- raises an exception if there is a lack of system entropy or if **trials** is set and it runs more than **trials** times, or
- returns `out` where `out` is $\top$ with probability **prob**, otherwise $\perp$.

If **trials** is set, the implementation's runtime is constant.

Proof. An integer sample is taken uniformly at random from $[0, \text{denom})$, where **denom** is the denominator of **prob**. The implementation then returns $\top$ if the sample is less than the numerator of **prob**, and $\perp$ otherwise. Since only at most **numer** outcomes of $\top$ are possible, out of **denom** possible outcomes, the implementation returns $\top$ with probability **prob**.

The implementation runs in constant-time if **trials** is set. □